

Multiple Choice (10 marks)

1. What characteristic is not representative of a type IIb muscle fibre?
 - (a) Low motor unit strength
 - (b) Low glycogen storage capacity
 - (c) High fatigue resistance
 - (d) Low oxidative capacity
 - (e) Low glycolytic capacity
2. Which of the following is NOT one of the major reasons for the increase in numbers of older adults?
 - (a) Increases in life expectancy at a specified age
 - (b) The increase in the rate of technological advancements
 - (c) The decrease in childhood mortality rates
 - (d) The steadily increasing birth rate
 - (e) The Baby Boom generation
3. The synthesis of glucose from lactate, glycerol, or amino acids is called:
 - (a) lipogenesis.
 - (b) lipolysis.
 - (c) lipo-genesis.
 - (d) gluconeogenesis.
 - (e) proteolysis.
4. The WHO made a strong recommendation that the maximum intake of free sugars by individuals within populations should be
 - (a) 10-15%
 - (b) >20%
 - (c) <10%
 - (d) >15%
 - (e) <5%
5. Which of the following is not a non-coding RNA?
 - (a) lncRNA
 - (b) piRNA

- (c) rRNA
- (d) snRNA
- (e) tRNA

True / False (10 marks)

Answer True or False

6. True or False: In the brain stem, pathways for discriminative touch decussate in the pyramids.
7. True or False: The karyotype 47,XX,+13 is diagnostic of Down syndrome.
8. True or False: Damage to the oculomotor nerve can lead to medially directed strabismus due to impaired eye muscle function.
9. True or False: Elephants are known to have a longer life span than humans, often living up to 80 years.
10. True or False: Senescence refers to the gradual loss of physical abilities associated with biological aging.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of Vitamin A in calcium absorption from the small intestine, analyzing its mechanisms and implications for overall health and nutrition.
12. Discuss the embryological development of the hyoid bone, focusing on the contributions of the third and fourth pharyngeal arches, and analyze its significance in human anatomy.

End of Paper